



A systematic review of software usability studies

Kalpna Sagar¹ · Anju Saha¹

Received: 15 March 2017 / Accepted: 11 October 2017
© Bharati Vidyapeeth's Institute of Computer Applications and Management 2017

Abstract The aim of this review is to summarize, analyze various research studies and identify different research gaps regarding usability standards and models, usability evaluation methods, usability metric, usability at different phases of software development life cycle and application domains of usability. This systematic review of usability studies between 1990 and 2016 has been conducted and 150 studies are identified. We conclude that researchers have not reached at consensus w.r.t. software usability models. We identify that *Efficiency*, *Effectiveness*, *Satisfaction*, and *Learnability* are commonly addressed attributes in various existing software usability models and standards. Further, developers do not have sufficient knowledge to decide the appropriate usability evaluation method to use in given domain. On the contrary, *Usability Testing*, *Heuristic Evaluation* and *Questionnaire* are identified frequently used methods for usability evaluation. Our findings investigate different metrics and measurement approaches used for usability estimation. But, current methods for usability measurement in practice do not include all ISO and ANSI defined aspects of usability into a single metric. Although, we identify studies concerning the integration of usability and software engineering into a single framework with generalizable results, their practical implementation is still missing and significantly needed. Conversely, this study highlights the fact that around 71% of studies address usability related issues during *Design-*

Phase of software development life cycle. At present, usability issues have been identified in various domains but around 33.82% of studies identify that usability evaluation approach is widely used in *Web-Domain*.

Keywords Software usability · Systematic literature review · Software development life cycle · Single usability metric · Object oriented metric · Master Usability Scaling

1 Introduction

The concept of usability emerges from the term *user-friendly*. Software usability is defined as the ease of use of the software. IEEE Std.610.12 explains usability as “the ease with which a user can learn to operate, prepare inputs for, and interpret outputs of a system or component” [1]. Usability is a vital aspect of software quality. Software quality can be assessed by a number of internal and external criteria's. A software quality model explains 6-broad categories of software quality: *functionality*, *reliability*, *usability*, *efficiency*, *maintainability*, and *portability*. In Fig. 1. these characteristics are further divided into sub-characteristics that have measurable attributes.

In software engineering, *usability* is one of most important software quality attribute. ISO/IEC 9126 describes usability as “a set of attributes that bear on the effort needed for use, and on the individual assessment of such use, by a stated or implied the set of users” [2]. The ISO/IEC 9126 redefines usability–“as the capability of the software to be understood, learned, used and liked by the user, when used under specified conditions”. According to ISO/IEC 9126 software quality model [2], usability has 5 attributes: *Understandability*, *Learnability*, *Operability*, *Attractiveness* and *Usability Compliance*. The term

✉ Kalpna Sagar
sagarkalpna87@gmail.com
Anju Saha
Anju_kochhar@yahoo.com

¹ University School of Information and Communication
Technology GGSIPU, Sector 16-c, Delhi 110078, India

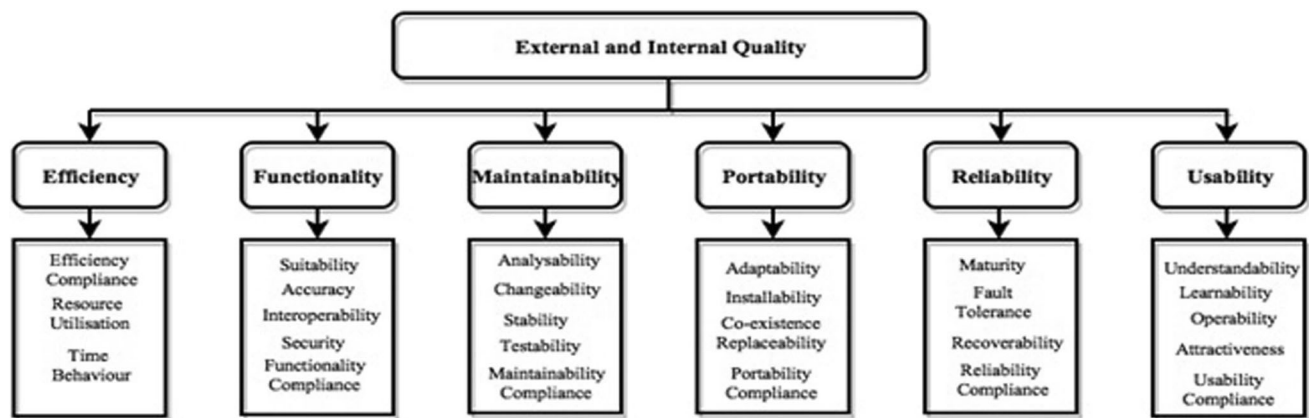


Fig. 1 ISO/IEC 9126 software quality model [2]

usability is defined in inconsistent manner by various standards and research practitioners. To demonstrate this point, few definitions of usability from some highly cited standards are mentioned below:

Nielsen [3], a pioneer from the field of software usability, defines usability as a set of 5-attributes that are explained below:

1. *Efficiency*: Systems should be efficient to use. When a user has fully learned the system, productivity will be on a high level.
2. *Errors*: The system should have a low error rate, which enables users to make few errors during the use of the system,
3. *Learnability*: Systems should be easy to learn. Users can rapidly start getting some work done with the system.
4. *Memorability*: Systems should be easy to remember, making it possible for casual users to return to the system after some period of not using the system, without having to learn everything all over again.
5. *Satisfaction*: The system should be pleasant to use, which makes users subjectively satisfied when using it.

ISO 9241-11 defines usability as “the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use” [4, 5]. ISO 9241-11 defines usability’s 3-attributes that are explained below:

1. *Effectiveness*: The accuracy and completeness with which users achieve specified goals.
2. *Efficiency*: The resources expended in relation to the accuracy and completeness with which users achieve goals.
3. *Satisfaction*: The comfort and acceptability of use.

In software engineering, usability is considered as one of most important software quality attribute. Many research

studies have revealed that usability problems indirectly or directly have caused software quality related problems. Usability related problems, mainly in interface design lead to lower efficiency and effectiveness. Due to this, end users face difficulties during its use [6]. In real world practice, poor usability is one of the main causes for failure of software application and product [7]. E-commerce studies revealed that a 5% improvement in usability could increase revenues by 10–35% [8]. Hence, there are significant benefits of usability for software companies and users (Nielsen). It helps in improving user productivity, product acceptability, reliability, user satisfaction, team morale, and e-commerce potential. It further produces financial benefits to companies, reduces training and documentation cost.

Usability engineering has various applications in various domains. It is considered to be wide research area in various countries like US, Scandinavian (Denmark) [9] etc. Interface design related issues in the information system, computer-based medical devices [10], mobile devices [11, 12], aviation system [13], banking system [14, 15] and radiotherapy treatment delivery systems [16] can be addressed through usability evaluation methods. Safety issues are also addressed by usability engineering. In the field of medicine, usability engineering is playing the significant role in improving the safety of health care information system [17–20], clinical information system [21, 22] and safety of patients [23].

Some contributors have focussed on the areas of usability and user experience [24, 25]. These relatively recent contributions have extended the dimensions of this open research area of usability to user experience. Additional psychometric work has considered more utilitarian aspects of usability to hedonics. This research has gone hand-in-hand with an evolution of design and development practices with a focus on usability to a more general appreciation of the importance of the entire user

experience. Kortum et al. has given significant information that must be of potential value to user experience researchers working for mobile applications [26–28]. Usability related issues are also evaluated for web-domain [29–33], input devices [34] and for mobile robot teleoperation [35]. An attempt has been made for integration of usability work into inter organisational agile development projects [36]. Furthermore, Software usability evaluation methods have also been adopted using data-mining techniques [37–39], and within model driven development approach [40].

The aim of this review is to summarize, analyze and identify different research gaps regarding usability standards and models, usability evaluation methods, usability metric, usability at different phases of software development life cycle and application domains of usability. This review is carried out as per guidelines given by Kitchenham [41]. This paper reviews all the studies done between the periods 1990 and 2016. In order to achieve this aim, we extensively searched through 7-electronic online databases¹ and identified 150-studies to answer the research questions pertaining to the software usability. The primary studies were selected based on the quality assessment and relevance.

The remaining part of this paper is organized as follows: Sect. 2 presents a method used to carry out this SLR. This section presents various research questions addressed in this study. It also includes adopted research criteria for selection of relevant primary studies. Section 3 presents answers to research questions identified in Sect. 2. Section 4 presents limitations of this study. Section 5 gives discussion, conclusions and future guidelines for this wide research area.

2 Method

In this paper, the Systematic Literature Review is carried out as per guidelines given by Kitchenham [41]. The process is clearly depicted in Fig. 2. Three phases i.e. planning, conducting and reporting the review results have been followed to carry out SLR. In the first phase, we identify the need for SLR and a review protocol is established to reduce the risk of research bias. Once the review protocol is approved, we execute the second phase i.e. conducting the review. In step 2.1, we define research questions addressing varied existing usability-related issues that are answered in this study. In step 2.2, a search strategy is defined that includes identification of various search terms and sources. By using these search terms, the

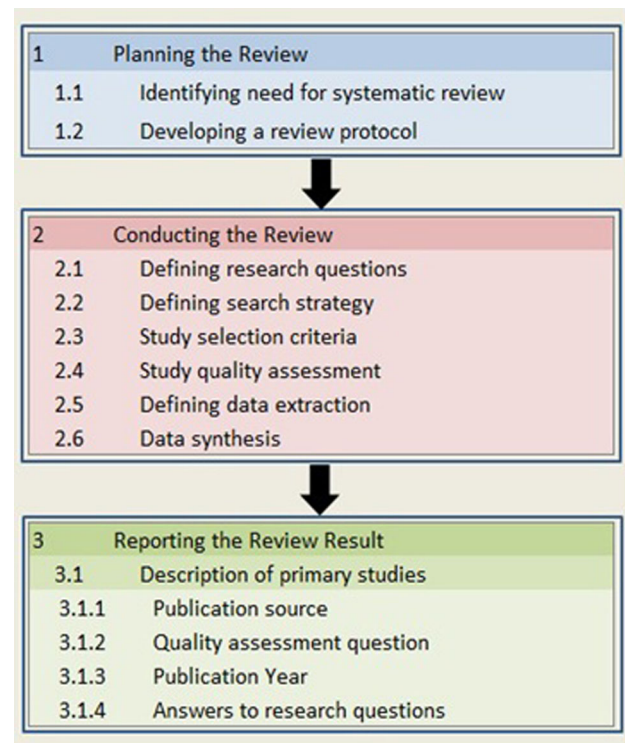


Fig. 2 Systematic literature review process

search is made to get primary studies related to software usability studies. In step 2.3, determination of relevant primary studies is done on the basis of inclusion–exclusion criteria’s and research questions defined in step 2.1. In step 2.4, we decide quality assessment questions to assess relevant primary studies. In step 2.5, we collect the required information to answer the research questions defined in step 2.1. In step 2.6, we devise methods for data synthesis after which the review results are attained and reported.

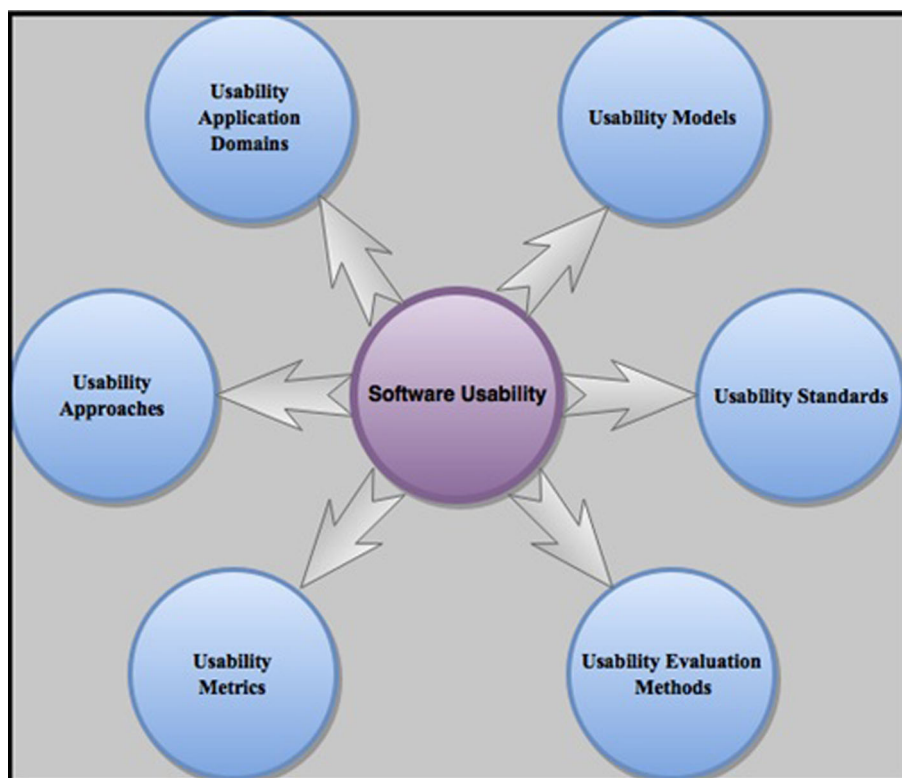
3 Planning the review

3.1 Identifying the need for systematic review

Although, there have been a large number of studies identified on software usability engineering. But, there is a lack of comprehensive study for this research area [42]. To provide a comprehensive study that explains software usability models, standards, evaluation methods, metric, and approaches etc., we need to conduct a systematic review. The review summarises all existing information about software usability in an unbiased way Fig. 3. represents different research areas of software usability that are explored in this paper.

¹ These online databases are mentioned in Sect. 2.2.

Fig. 3 Software usability research areas



3.2 Developing a review protocol

In this phase, review protocol is established to reduce the risk of research bias. The review protocol is developed by having meetings in a group of various Associate Professors.²

4 Conducting the review

4.1 Defining research questions

The goal of this literature review is to provide the empirical evidence obtained from various studies of software usability engineering. The study presents 5-different research questions that are addressed in Table 1. In RQ1, various software usability models and standards are listed. We analyze these studies to answer RQ1.1. The aim of this question is to identify frequently addressed usability attributes in various existing software usability models and standards. In RQ2, various methods used for evaluation of software usability, are presented. After analysis of selected primary studies, we get an answer to RQ2.1. The aim of this question is to determine frequently used approach for

evaluation of software usability. In RQ3, different metrics used for estimation or measurement of software usability, are also identified. In RQ4, frameworks for the integration of usability engineering with software engineering are explored. From selected primary studies, we get answer to RQ4.1. The purpose of this question is to know about various phases of software development life cycle where usability related issues are addressed the most. In RQ5 determines the various domains of application for software usability related issues.

4.2 Defining search strategy

Search terms are formed by having alternative terms and synonyms using Boolean expression 'OR' and by combining main search terms using 'AND' Table 2. presents general search terms used for identification of primary studies related to software usability.

Once the search terms are identified, the significant digital portals are selected. For searching the primary studies, we explore following 7-electronic databases: SpringerLink, IEEE Xplore, ACM Digital Library, ELSEVIER, Taylor and Francis online, ScienceDirect and Google scholar. In addition to these, we also search in some international journals and conferences.³

² Associate professors are from University School of Information and Communication Technology, Guru Gobind Singh Indraprastha University, Delhi, India.

³ List of publication sources from international journals and conferences is provided in Table 4.

Table 1 Research questions

RQ#	Research questions	Motivation
RQ1	Which usability models and standards have been used in the past for analysis of software usability?	Identify various models and standards for software usability
RQ1.1	Which attributes of software usability are most commonly addressed in various usability models and standards?	Identify frequently used attributes for usability in usability models and standards
RQ2	Which methods have been introduced for the evaluation of software usability?	Identify various software usability evaluation methods
RQ2.1	Which method is most commonly used for the evaluation of software usability?	Identify various software usability evaluation method
RQ3	What are various metrics used for estimation of software usability?	Identify metrics (or measurement tools) for estimation of software usability
RQ4	Can we integrate usability engineering with software engineering?	Identify framework for integration of usability engineering with software engineering
RQ4.1	At which phase of software development life cycle, software usability issues are addressed the most?	Identify the phases of software development life cycle to address usability related issues
RQ5	What are the various domains of application for usability?	Investigate domains of applications of usability

Table 2 Search terms used for identification of primary studies

Search terms
Software AND (usability) AND (models OR standards)
Usability AND evaluation methods (OR Inspection Methods OR Testing Methods OR Inquiry Methods)
Software AND (usability) AND (metrics OR measurement OR estimation OR evaluation OR assess)
Integration of usability engineering into software engineering (OR Usability with AND((requirement phase OR design phase OR implementation phase))
Application of usability in AND (various domains OR Software)

4.3 Study selection criteria

The first software usability model has been introduced in 1977. But, this study has focused on usability studies from 1990 to 2016. Relevant research areas of software usability studies are classified into 5-major categories. Furthermore, relevant primary studies are identified by obtaining full-text papers on the basis of below-mentioned inclusion and exclusion criteria's. The paper has also included some full-text papers found in the reference section of searched primary studies. The SLR considers empirical and analytical primary studies for inclusion in the reference list of primary studies. Hence, for further processing, total numbers of 172 studies are identified. Some of the studies are excluded because they do not give answers to any of identified 5-research questions and quality assessment questions of step.2.4. 150-primary studies are selected on the basis of following inclusion–exclusion criteria's:

Inclusion criteria:

1. Empirical and Analytical studies for software usability models and standards (with the taxonomic division of usability's attributes).
2. Studies using usability evaluation methods.

3. Empirical studies contributing for usability metrics.
4. Studies contributing towards integration of usability engineering with software engineering.
5. Empirical studies for applications of usability for different domains.

Exclusion criteria:

1. Studies using techniques in the context other than addressing usability related issues.
2. Studies without empirical analysis of usability evaluation methods.
3. Similar Studies with same results and by same authors but in different conferences and journals.
4. Studies written in a language other than English.

The above-mentioned criteria's are tested by 2-researchers⁴ individually and they have given a common decision after detailed study. Furthermore, quality of selected primary studies is determined by their relevance to defined research questions. In next subsection, quality

⁴ 2-researchers are Kalpna Sagar, a PhD-Student and Dr. Anju Saha, an Associate Professor in University School of Information and Communication Technology, GGSIPU, Delhi, India.

assessment criteria's are presented to get final studies in this SLR.

4.4 Study quality assessment

In order to find the relevance of primary studies, quality assessment questions in Table 3 are presented. The purpose of these questions is as a means of weighting the importance of individual selected primary studies when the results are being synthesized. We have considered suggestion given in [42] and [41] for the development of these questions. Ranking of the questions as 1 (yes), 0 (no) and 0.5 (partly) is also done. Final score for each selected primary study after adding the values assigned to each quality assessment question are obtained. A study can have a maximum score of 12 and a minimum score of 0. 2-Researchers play a major role in ranking the quality questions and in a case of disagreement, consultation with others researchers have been taken. After which, a robust and final decision is taken for each primary study.

4.5 Defining data extraction

For each selected primary studies, a data extraction form is prepared and filled to identify which research questions are satisfied by the primary study. The basic objective of this step is to record information from primary studies. For this, details like title, author name, journal publication, year of publication, usability models, standards, attributes, evaluation methods, metrics, integration of usability at different phases of software development life cycle and various domains for the application of usability, are summarised. Another researcher is also asked to collect information

required on a random sample of primary studies. The aim of this step is to cross-check the results. In a case of disagreement with results, other researchers are asked to resolve a disagreement. The resultant data is saved into word file (Tabular form), so that if required then resultant data can be used further for data synthesis step.

4.6 Data synthesis

The aim of this step is to collate and summarise results from selected primary studies [41]. In this study, data synthesis step is non-quantitative and descriptive. The paper has tabulated extracted information from selected primary studies in the consistent manner so that we can get answers to the research questions. In order to synthesize data from primary studies, we have used visualization techniques from ms-excel e.g. column charts, line graph and pie charts etc. Tables from ms-office have also been used for presenting results.

5 Reporting the Review Results

This portion explains results of selected primary studies. Based on the extensive research on studies, some of the notable key points are extracted. First of all, sub-sections present description of each primary study with their publication sources. After this, scores for each primary studies are evaluated. This score evaluation is based on guidelines mentioned in Tables 3 and 5. The study provides year-wise distribution of selected research publications. Answers to each research questions are also discussed in sub-sections.

Table 3 Quality assessment questions

Quality assessment questions	1 (yes)	0 (no)	0.5 (partly)
1. Are the aims of the research clearly stated?			
2. Are different attributes used to assess usability in usability models/standards clearly defined?			
3. Are usability evaluation methods/sub-methods clearly defined with detailed description?			
4. Does study provide guidelines on how usability evaluation methods can be applied for evaluation of usability in various fields?			
5. Are metrics used for estimation of usability?			
6. Are usability related issues clearly addressed and managed during SDLC?			
7. Does study highlight domain of applications of usability related issues?			
8. Are the results and findings clearly stated in study?			
9. Are the limitations of study specified?			
10. Is the research methodology repeatable?			
11. Does study contribute to literature?			
12. Does study cited by other authors?			

* 1 (yes)—when author finds answer to stated questions, 0 (no)—when author does not find answer, 0.5 (partly)—when author partly gets answer

Table 4 Summary of selected primary studies in different publications

Publication name	Type	Number	Percentage (%)
ACM Computing Surveys	SCI-J	1	0.66
ACM Transactions on Computer–Human Interaction	SCI-J	1	0.66
ACM Transactions on Information Systems	SCI-J	1	0.66
Advances in Engineering Software	SCI-J	2	1.33
Applied Ergonomics	SCI-J	1	0.66
Communications of the ACM	SCI-J	6	4
Decision Support Systems	SCI-J	1	0.66
Ergonomics	SCI-J	1	0.66
Human–Computer Interaction	SCI-J	1	0.66
IEEE Transactions on Software Engineering	SCI-J	2	1.33
Information and Software Technology	SCI-J	7	4.66
International Journal of Design	SCI-J	1	0.66
International Journal of Electronic Commerce	SCI-J	1	0.66
International Journal of Human–Computer Interaction	SCI-J	21	14
International Journal of Human Computer Studies	SCI-J	23	15.33
International Journal of Industrial Ergonomics	SCI-J	1	0.66
International Journal of Medical Informatics	SCI-J	3	2
Journal of Medical Internet Research	SCI-J	1	0.66
Journal of Medical Systems	SCI-J	1	0.66
Journal of Systems and Software	SCI-J	10	6.66
Software Quality Journal	SCI-J	1	0.66
Springerplus	SCI-J	1	0.66
Radiotherapy and Oncology	SCI-J	1	0.66
Others Journals*	Non-SCI	29	19.33
International Conferences	Conferences	32	21.33
Total		150	

* In other journals, we have considered open access journals e.g. E-Informatica Software Engineering Journal, Behaviour and Information Technology, Journal of Software Engineering Research and Development and Universal Access in Information Society etc

5.1 Description of primary studies

This part gives the brief description about selected primary studies. For this SLR, we have included 150 primary studies, out of 172 studies. A complete list⁵ of selected primary studies is presented in this SLR. We also assign final scores⁶ to each selected primary study after adding the values assigned to each quality assessment questions.⁷

5.2 Publication source

This section summarizes the details of research publications that are selected for review. These publications are found in reputed international journals and conferences from 7-electronic databases¹. Table 4 contains the number

and percentage of included primary studies. The major publications are found in International Journal of Human Computer Studies, International Journal of Human–Computer Interaction, Journal of System and Software, Information and Software Technology, Communications of the ACM and so on. From Table 4, the statistics show that 78.66% of primary studies are presented in *international journals*⁸ whereas 21.33% of primary studies are presented in *international conferences* (included 32 international conferences in total). Hence, the number of primary studies which has published in international journals is more than international conferences. Table 4. Further shows that 15.33% of primary studies are published in *International Journal of Human Computer Studies* and 14% in *International Journal of Human–Computer Interaction* and 6.66% in *Journal of systems and software* and so on.

⁵ This list is provided in Table 6.

⁶ Final scores details are presented in Table 5.

⁷ These questions are mentioned in Table 3.

⁸ We have included 89 research publications from SCI-Indexed journals and 29 from Non-SCI journals.

Table 5 Scores details based on Table 3

Categories of scores	Range of scores	Scores	Included primary studies
High	$(6 \leq \text{score} \leq 12)$	9	S133
		8.5	S97, S117 and S134–S135
		8	S26, S34, S42, S62, S79, S85, S93, S96, S122 and S138
		7.5	S30, S41, S65, S90, S98, S102, S115, S120, S129, S137 and S144
		7	S6, S17, S39, S67, S75, S82, S89, S95, S107, S111, S114, S121, S126, S128, S146 and S148
		6.5	S8, S10–S12, S21, S23, S28–S29, S33, S36, S44–S46, S63, S66, S83–S84, S100–S101, S108, S110, S112, S119, S132, S136, S140–S143 and S150
Medium	$(3 \leq \text{score} < 6)$	6	S5, S16, S19, S22, S37, S40, S52–S53, S58, S113, S116, S123, S125, S131, S145 and S147
		5.5	S4, S15, S20, S27, S47–S48, S50–S51, S57, S60, S69, S73, S77–S78, S86, S91, S99, S118, S130 and S139
		5	S3, S7, S18, S35, S38, S61, S68, S71, S74, S80–S81, S88, S92, S94, S106, S127 and S149
		4.5	S9, S25, S31–S32, S56, S72, S87, S109 and S124
		4	S1–S2, S13–S14, S24, S49, S54–S55, S76 and S103–S104
		3.5	S64 and S70
Low	$(0 \leq \text{score} < 3)$	3	S59 and S105
		2.5	S43
		2–0	–

* Maximum value: 12 and minimum value: 0

5.3 Quality assessment questions

Final scores⁹ are evaluated for each included primary studies. Categories, the range of scores and description of included primary studies with their corresponding scores are given in Table 5.

Based on the extensive research on quality assessment questions, here, the study highlights some of the noteworthy points:

1. The study recognizes that some of the papers may not have yet obtained citations, we assess 15-papers (S28, S33, S39, S43, S64, S72, S106, S113, S114, S130, S139, S142, S146, and S149–S150) excluding quality question Q12.
2. 17-studies S4, S6, S24, S44, S46–S47, S51, S55, S60, S77–S78, S82, S87, S88, S89, S94, and S107 are the high rankers in terms of citations by other authors. It is also worth mentioning that maximum of these papers has been published in the ACM.
3. 88-studies S5–S6, S8, S10–S12, S16–S17, S19, S21–S23, S26, S28–S30, S33–S34, S36–S37, S39–S42, S44–S46, S52–S53, S58, S62–S63, S65–S67, S75, S79, S82–S85, S89–S90, S93, S95–S98, S100–S102, S107–S108, S110–S112, S113–S117, S119–S122, S123, S125–S126, S128–S129, S131–S138, S140–S148, and S150 have obtained high score. Hence, the

interested research practitioners may take these publications into consideration for further reading.

4. Studies S1, S3, S6–S7, S18, S42–S43, S45–S48, S59, S97–S98, S107–S109, S116, S134, S135, and S144 cite existing different *software usability models and standards*.
5. Studies S4, S12, S20, S27, S41, S44, S49, S53, S61, S72–S73, S85–S86, S90, S92, S106, S118, S120, S122–S123, S125–S127, S130–S131, S138–S139, and S150 refer *usability testing* methods.
6. Studies S3, S12, S30, S36–S37, S40, S44, S62–S63, S65, S79, S86, S88–S90, S114, S117, S120, S122–S123, S126, S138, and S142 mention *heuristic evaluation* methods.
7. Studies S16, S21, S27, S32, S34–S35, S38, S72, S83, S90, S96–S97, S99, S101, S115–S116, S122, S126, S128–S129, S133, S135–S136, S138, S140–S142, S144, and S146–S150 refer *questionnaire* methods.
8. Studies S4, S27, S29, S42, S44–S52, S54–S57, S59, S62, S64, S72, S79, S83, S96, S98, S107, S111, S113, S129, S134, S136, S140, and S147 mention various *metrics or measurement* approaches for estimation of usability.
9. Studies S4–S6, S8–S12, S15–S17, S22–S23, S26, S28, S39–S42, S45, S50, S53, S65, S66, S76, S82, S84, S90, S93, S97–S101, S106, S110, S114–S117, S122–S124, S126, S129, S133, S138–S139, S141–S144, S146, S148, and S150 cite for *design* phase where usability related issues are addressed the most.

⁹ Final score is computed for each selected primary study after adding the values assigned to each quality assessment question⁷.

10. Studies S34, S49, S58, S61–S63, S76, S79, S85, S96–S98, S100, S106, S112, S117, S120, S130, S133, S142–S143, S145, and S148 refer *web-domain* where the application of usability is found most significant.

The studies which do not answer any of identified research question¹⁰ are excluded from the analysis. Hence, after quality assessment, further 16-studies are removed from the analysis.

In Table 6, a unique identifier (i.e. Study nos) is assigned to each selected primary study with their corresponding reference number. The selected primary studies would be referred by their corresponding unique identifier if required in subsections Table 6. Also, provides details of authors of selected primary studies with their publication year.

5.4 Publication year

Figure 4 represents the distribution of selected primary studies from 1990 to 2016. It shows that before 2004 out of 150 only 26 studies addressed usability related research areas. Most of the studies are conducted from 2004 onwards. The number of studies in the years 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013, 2014, 2015 and 2016, are 12, 11, 9, 7, 9, 8, 6, 11, 6, 14, 6, 18 and 7 respectively accounting about 82.66% of studies between 2004 and 2016. Since the year 2004, the studies have started examining the effectiveness of usability related research areas. This confirms the inclusion of most relevant studies in this SLR. Figure 4. further reflects that year 2015 has maximum research publications for software usability, accounting 12%. Hence, the study concludes that software usability is an open research area.

2.1.1 RQ1: Which usability models and standards have been used for analysis of software usability?

Over the years, a number of usability models have been introduced and used for analysis of software usability. Table 7 gives the details of different usability models with author's names and year in which that model has been introduced. Table 7 also gives the description of usability's attributes, defined by these authors. Studies which are included in this category are S1, S3, S6–S7, S18, S42–S43, S45–S48, S97, S98 (usability model relates human interface elements to usability dimensions using multivariate statistical techniques), S107–S109, and S116, S134, S135, S144. The study S59 describes evaluation model that considers usability with security. The Study S43 provides usability risk assessment model. Moreover, we have

searched Google scholar and search engine for further exploration of usability models.

The study represents various usability models proposed by various research practitioners and addresses a research gap belonging to these models. It has been identified that researchers have not reached at consensus w.r.t. usability models. So, there is a need of consolidated model of usability with non-redundant taxonomic division of its attributes. On the contrary, the purpose, uses, and benefits of usability models have not been provided in any of primary study. Further, validation of the proposed models is also missing.

Table 8 gives the description of different usability standards with their publication year and description. The summary of usability related standards has been presented to guide the novice researchers. Studies which are included in this category are S6, S7, S45, S48, and S79. The study S47 describes the working model which is an improvement over ISO 9241-11.

- **RQ1.1:** Which attributes of software usability are frequently addressed among different usability models and standards?

This section gives the details of most commonly addressed attributes of software usability among different existing usability models and standards. Figure 5 represents frequency count for usability attributes in existing usability models and standards. Figure 5 also concludes that *Learnability*, *Efficiency*, *Satisfaction*, and *Effectiveness* are frequently addressed usability attributes.

Aesthetics, Communicativeness, Errors for its target users, Functional Design, Helpfulness, Interface Design, Productivity, Robustness, Retainability, Subjectively pleasing, Security, Type of Users, Trustfulness, Training Material and Universality are addressed twice in existing usability models and standards.

Furthermore, Acceptability, Attractiveness, Adaptiveness, Appropriateness recognisability, Computer Systems and Networks, Control, Components, Consistency, Context, Data and Metadata, Ease of Use, Effort for error correction, Engagement, Functionally Correct, First Impression, Feature Usage, Human Factors, Information Quality, Interface Quality, Knowability, Modifiability, Maintainability, Outcome, Portability, Process, Rate of Error, Reusability, Reliability, Reliability in use, Relevance, Social, Scalability, System Function, Speed of performance, Throughput, Time to learn, Type of Product, Technical, User compliance, Use, User documentation and wizards and agents are addressed only once in existing usability models and standards.

- **RQ2:** Which methods have been introduced for the evaluation of software usability?

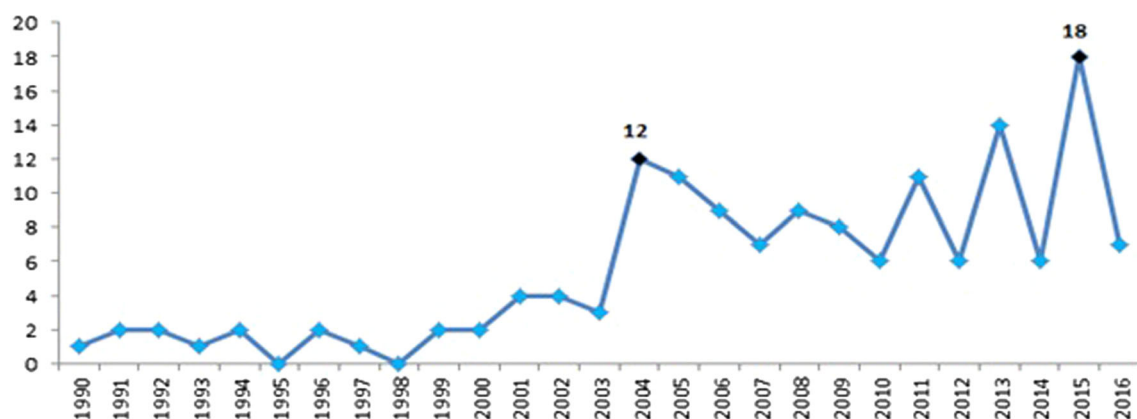
¹⁰ 5-Research questions are presented in Table 1.

Table 6 Selected primary studies (S1 to S150)

Study no.	Author	Year	Ref no.	Study no.	Author	Year	Ref no.
S1	N. Bevan	1997	[43]	S31	T. Clemmensen	2011	[44]
S2	H. J. Heiskari	2009	[45]	S32	Winter. J	2009	[46]
S3	C. Lallemand	2011	[47]	S33	Polgár. B. P	2015	[48]
S4	J. Burr	1999	[49]	S34	Y. Lee	2012	[50]
S5	L. Bass	2003	[51]	S35	S. G. Vilbergdottira	2014	[52]
S6	E. Folmer	2004	[53]	S36	Graham. M. J	2004	[10]
S7	K. Nebe	2009	[54]	S37	Chan. A. J	2012	[16]
S8	E. Folmer	2005	[55]	S38	S. Makri	2011	[56]
S9	B. E. John	2004	[57]	S39	Hiniker. A	2016	[58]
S10	N. Juristo	2007	[59]	S40	M. A. Khawaja	2014	[60]
S11	A. Seffah	2008	[61]	S41	B. Aryana	2013	[62]
S12	N. Juristo	2007	[63]	S42	D. E. Tamir	2010	[64]
S13	A. Seffah	2001	[65]	S43	J. T. S. Moorthy	2013	[66]
S14	A. Beu	2011	[67]	S44	J. Nielsen	1993	[68]
S15	S. Bodker	2002	[12]	S45	S. K. Dubey	2011	[69]
S16	M. J. Roberts	2013	[70]	S46	J. Sauro	2005	[71]
S17	M. Hertzum	2006	[72]	S47	K. Hornbaek	2006	[73]
S18	E. Metzker	2001	[74]	S48	M. F. Bertoa	2006	[8]
S19	M. Hertzum	2012	[75]	S49	W. Erickson	2013	[76]
S20	J. Jankowski	2015	[35]	S50	S. K. Dubey	2010	[77]
S21	T. Lavie	2011	[78]	S51	E. Frojkaer	2000	[79]
S22	L. Carvajal	2013	[80]	S52	E. M. Babiker	1991	[81]
S23	L. Carvajal	2009	[82]	S53	B. Beckles	2005	[83]
S24	A. Seffah	2004	[7]	S54	M. Mcgee	2004	[84]
S25	R. Schusteritsch	2007	[11]	S55	J. Nielsen	1994	[85]
S26	A. W. Kushniruka	2005	[17]	S56	J. Sauro	2005	[86]
S27	M. A. Teruel	2014	[87]	S57	J. Kjeldskov	2004	[21]
S28	G. Lindgaard	2015	[88]	S58	M. B. Skov	2005	[89]
S29	P. Kortum	2015	[26]	S59	T. Mapayi	2013	[90]
S30	G. H. J. Johannessen	2014	[91]	S60	M. Y. Ivory	2001	[92]
S61	S. Lee	2007	[93]	S106	F. D. Rodriguez	2015	[30]
S62	M. Gonzalez	2009	[94]	S107	A. Seffah	2006	[6]
S63	Delice	2009	[95]	S108	D. Alonso-Ríos	2010	[96]
S64	E. Bradner	2008	[97]	S109	S. Winter	2008	[98]
S65	B. R. Kline	2005	[99]	S110	A. Y. W. Kolade	2015	[36]
S66	D. Tamir	2008	[100]	S111	T. Scheller	2015	[101]
S67	T. Tulis	2002	[102]	S112	V. F. D. Santana	2015	[103]
S68	A. Folstad	2007	[104]	S113	H. Hoehle	2016	[27]
S69	E. Frokjaer	2005	[105]	S114	M. Brown	2013	[106]
S70	K. E. Thompson	2004	[107]	S115	J. I. Panach	2015	[40]
S71	J. O. Bak	2008	[108]	S116	R. Harrison	2013	[109]
S72	J. Winter	2011	[110]	S117	D. Castilla	2013	[111]
S73	A. Bruun	2012	[112]	S118	S. L. Mansar	2012	[19]
S74	J. Sauer	2010	[113]	S119	N. Hurtado	2015	[114]
S75	L. Hua	2013	[22]	S120	L. Hasan	2012	[115]
S76	J. I. Hong	2001	[116]	S121	J. Kjeldskov	2004	[117]
S77	J. Nielsen	1990	[118]	S122	Y. G. Ji	2006	[119]
S78	J. Nielsen	1992	[120]	S123	M. Clamann	2004	[13]

Table 6 continued

Study no.	Author	Year	Ref no.	Study no.	Author	Year	Ref no.
S79	M. C. S. Torrente	2013	[121]	S124	E. Folmer	2006	[122]
S80	M. S. Andreasen	2007	[123]	S125	V. L. Roberts	2006	[124]
S81	P. C. Wright	1991	[125]	S126	A. M. Nivala	2007	[126]
S82	C. Wharton	1994	[127]	S127	S. Propp	2009	[128]
S83	M. C. Leow	2016	[129]	S128	G. Peevers	2008	[14]
S84	A. Folstada	2010	[130]	S129	C. S. Weir	2006	[15]
S85	M. B. Skov	2004	[131]	S130	W. A. M. Isa	2011	[132]
S86	M. P. Gonzalez	2008	[37]	S131	P. Yammiyavar	2008	[133]
S87	H. R. Hartson	1996	[134]	S132	M. Hertzum	2011	[135]
S88	J. Nielsen	1992	[136]	S133	L. Rivero	2013	[137]
S89	J. Mankoff	2003	[138]	S134	B. C. Zapata	2015	[23]
S90	S. M. Ko	2013	[139]	S135	K. Moumane	2016	[28]
S91	P. Sawyer	1996	[140]	S136	K. C. Hamborg	2004	[141]
S92	J. M. C. Bastein	2010	[18]	S137	M. Arnhold	2014	[142]
S93	M. Paradowski	2004	[143]	S138	J. Mirkovic	2014	[144]
S94	A. Holzinger	2005	[145]	S139	M. A. McClellan	2016	[29]
S95	R. Spencer	2000	[146]	S140	M. A. Rupp	2015	[34]
S96	T. Christophersen	2011	[147]	S141	D. Datcu	2015	[148]
S97	S. Leuthold	2008	[149]	S142	R. G. Bias	2015	[150]
S98	K. J. Kim	2002	[151]	S143	L. L. Tung	2014	[152]
S99	W. Chung	2005	[153]	S144	I. K. Choi	2015	[154]
S100	V. Roth	2005	[155]	S145	A. Nawaz	2013	[156]
S101	M. Howell	2006	[157]	S146	M. Hochhauser	2015	[20]
S102	A. Fernandez	2011	[42]	S147	J. R. Lewis	2015	[25]
S103	T. Borgholm	1999	[9]	S148	R. P. Bringula	2016	[31]
S104	R. West	2006	[158]	S149	A. Shamim	2016	[159]
S105	D. Wixon	2003	[160]	S150	S. Hanrath	2015	[161]

**Fig. 4** Year wise distribution of primary studies

This part gives the details of methods introduced for evaluation of software usability. The study S79 describes *ISO DTR 16982 (2001)* that introduces different evaluation

methods (www.iso.org). Studies which are included in this category are S3–S4, S6–S8, S11–S12, S14, S16, S17, S19–S21, S26–S34, S36–S37, S41, S44, S46, S49, S53, S57–

Table 7 Usability models (1977–2016)

Usability Models	Year	Usability attributes
Abran et al. [162]	2003	Efficiency, effectiveness, satisfaction, learnability, security
Alonso-Rios et al. [96]	2009	Knowability, operability, efficiency, robustness, safety, subjective satisfaction
Arms [163]	2000	Data and metadata, interface design, functional design, computer systems and networks
Bass et al. [51]	2003	Modifiability, reusability, security, scalability, performance
Bevan et al. [164]	1991	Type of product, type of user, ease of use, acceptability
Boehm [165]	1978	Portability, maintainability
Booth [166]	1989	Usefulness, effectiveness, learnability, attitude
Battleson et al. [167]	2001	Remember, use, errors for its target users, specific tasks, easy to learn
Blandford et al. [168]	2002	Technical, cognitive, social, design-oriented
Brinck et al. [169]	2002	Efficient to use, learn, remember, error tolerant, subjectively pleasing, functionally correct
Constantine et al. [170]	1999	Efficiency in use, learnability, rememberability, reliability in use, user satisfaction
Tamir et al. [64]	2010	Learnability, operability, understandability
Dix et al. [171]	1998	Learnability, flexibility, robustness
Donyaee et al. [172]	2001	Efficiency, effectiveness, productivity, satisfaction, learnability, safety, trustfulness, accessibility, universality, usefulness
Frokjaer et al. [79]	2000	Components, effectiveness, efficiency, satisfaction
FURPS [173]	1992	Aesthetics, human factors, user documentation, consistency, wizards and agents, training materials, online and context sensitive help
GOULD [174]	1988	System performance, system functions, user interface
Hix et al. [175]	1993	Learnability, long-term user satisfaction, advanced feature usage, first impression, performance, retainability
Löwgren [176]	1993	Relevance, efficiency, learnability, attitude
Lewis [167]	1995	System usefulness, information quality, interface quality
McCall [177]	1977	Operability, training, communicativeness
MUSiC [178]	1997	User performance measures(task effectiveness, temporal Efficiency, length or proportion of productive period)
Nielsen [3]	1993	Learnability, efficiency, memorability, errors, satisfaction
Preece et al. [179]	1993	Safety, effectiveness, efficiency, enjoyableness
Preece et al. [180]	1994	Learnability, efficiency, throughput, flexibility, attitude
Quesenbery [181]	2001–2004	Effectiveness, efficiency, engagement, error tolerance, ease of learning
Harrison et al. PACMAD [109]	2013	Effectiveness, efficiency, satisfaction, learnability, memorability, errors, cognitive load, user, task, context
Seffah et al. QUIM [172]	2001	Efficiency, effectiveness, productivity, satisfaction, learnability, safety, trustfulness, accessibility, universality, usefulness
Schneiderman et al. [182]	1992	Rate of errors by users, subjective satisfaction, speed of performance, retention over time, time to learn
Shackel [183]	1991	Effectiveness, learnability, flexibility, subjectively pleasing

S58, S60–S63, S65, S71–S97, S99–S106, S110, S112, S114–S123, S125–S129, S130–S131.

- *RQ2.1*: Which method is most commonly used for the evaluation of software usability?

Over the years, different evaluation methods have been used for the evaluation of software usability. Table 9 gives the details of varied usability evaluation methods with included primary studies under the same.

Usability evaluation has been done using various methods mentioned in Table 9. The study identifies 37-different methods for usability evaluation. Out of all the

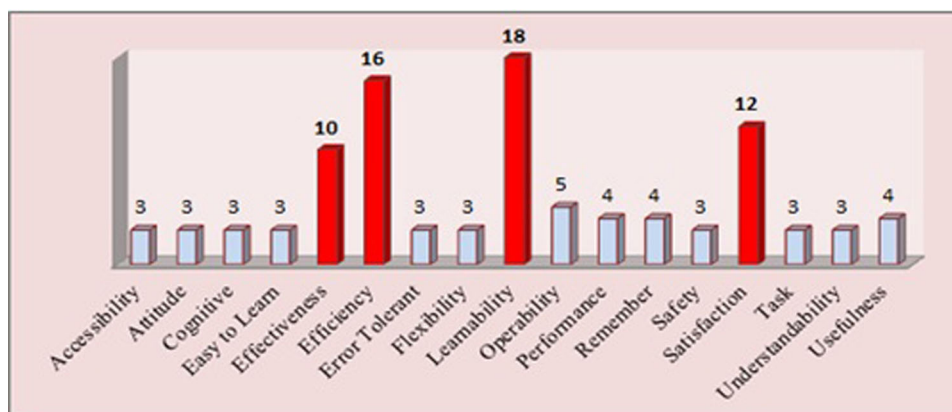
usability evaluation methods, the usability testing, heuristic evaluation, and questionnaire methods have been used extensively. But, no study is found based upon Coaching Method, Codiscovery Learning, Retrospective Testing, Retrospective Thinking-Aloud, Shadowing Method, Teaching, Consistency Inspection, Diagnostic Evaluation, Formal Usability Inspection, Participatory Heuristic Evaluation, and Participatory Walkthrough, Pluralistic Walkthrough, and Perspective-Based Inspection methods, to best of author's knowledge. Hence, the study represents a research gap related to usability evaluation methods and highlights those evaluation methods which have not been

Table 8 Usability-related standards (1990–2016) [1, 2, 184–188] (www.iso.org)

Usability standards	Year	Description
IEEE Std. 1061	1992	It describes comprehensibility, ease of learning, and communicativeness as usability attributes
ISO/IEC 9126	1991	It defines usability as a set of attributes that bear on the effort needed for use, and on the individual assessment of such use, by a stated or implied set of users
ISO 9241-11	1998	It explains usability as set of 3-attributes. These are Effectiveness, Efficiency, and Satisfaction
ISO 25010	2011	It specifies learnability, appropriateness recognisability, operability, user error protection, user interface aesthetics and accessibility as usability attributes
Use in contexts		
ISO/IEC DTR 9126-4	2001	It suggests metrics for effectiveness, productivity, satisfaction and safety
ISO WD 20282	2001	It specifies information about usability given with a consumer product, so that a purchaser can judge the ease of use of the product
Interface and interaction		
ISO 9241	1992–2000	ISO 9241 provides requirements and recommendations relating to the attributes of the hardware, software and environment that contribute to usability, and the ergonomic principles underlying them. Parts 3–9 contain hardware design requirements and guidance
ISO/IEC 10741-1	1995	It specifies how the cursor should move on the screen in response to the use of cursor control keys
ISO AWI 18789	1999	It contains requirements and measurement techniques for electronic visual displays
ISO/IEC 14754	1999	It defines a set of basic gesture commands and feedback for pen interfaces
ISO/IEC 15910	1999	It provides a detailed process for the development of user documentation (paper and on-line help)
ISO/IEC 11581	1999–2000	It contains requirements and recommendations for Icon symbols and functions
ISO 13406	1999–2001	It establishes ergonomic image-quality requirements for the design and evaluation of flat panel displays and specifies methods of determining image quality
ISO/IEC 9126-1	2000	It expresses usability in terms of Understandability, learnability, operability and attractiveness. It also specifies and evaluates corresponding metrics and criteria
ISO/IEC 18019	2000	It gives more guidance on how to produce documentation that meets user needs
ISO 14915	2000	It provides guidelines for multimedia user interfaces
IEC CDV TR 61997	2000	It gives general principles and detailed design guidance for media selection, and for mechanical, graphical and auditory user interfaces
ISO/IEC FCD 18021	2001	It contains user interface specifications for PDAs with a data interchange capability with corresponding servers
User centred process		
ISO/IEC 14598	1998–2008	It gives a general framework for the evaluation of software products using the model in ISO/IEC 9126-1
ISO 13407	1998	It explains the activities required for user centred design processes. interactive system
ISO TR 16982	2001	It outlines different types of usability evaluation methods that can be used to support human centred design
Usability capability of organisation		
ISO TR 18529	2000	It adds Usability maturity model containing a structured set of processes derived from ISO 13407 and a survey of good practice. UMM contains 7 groups of basic tasks explaining Human centered Design Processes and their activities and includes system users in its lifecycle
Other usability related standards		
ISO 20282-1	2006	It provides requirements and recommendations for the design of easy-to-operate everyday products. It addresses usability concerned with the user interface by taking account of the relevant user characteristics and the context of use
ISO 9241-151	2008	Web specific standard provides guidance on human centred design of software web user interfaces with aim of increasing usability. It mainly focuses on high-level design decisions and design strategy; content design; navigation and search; content presentation
ISO/TR 9241-100	2010	It enables users of standards related to software ergonomics to identify ergonomics standards relevant to software development of an interactive system affecting usability
ISO/IEC TR 25060	2010	It describes a potential family of International Standards, named the Common Industry Formats (CIF), that document the specification and evaluation of the usability of interactive systems
ISO/IEC 25010	2011	A quality model that replaces ISO 9126-1 but uses same definition as ISO 9241-11 with broader concept of quality in use (mainly security and compatibility are added)

Table 8 continued

Usability standards	Year	Description
IEC 62366-1	2015	Application of usability engineering to medical devices
IEC/TR 62366-2	2016	Guidance on the application of usability engineering to medical devices
ISO 9001	2000	It involves the systematic identification of requirements for usability, including usability measures and verifiable description of the resulting design. These provide design targets which can be basis for verification of the resulting design

**Fig. 5** Frequency count of generally addressed usability attributes in existing usability models and standards

used till yet. This is an open issue which is not addressed in any of study so far. Research practitioners must focus, explore these methods and should use them for future research. Moreover, the practitioner should ideally combine both qualitative (i.e. formative methods) and quantitative methods (i.e. summative methods) for evaluation of usability [189]. But no primary study has been identified that combine both evaluation methods. Hence, researchers must put their efforts and made an attempt to combine both evaluation methods during evaluation of usability.

There are also different tools available to support software usability evaluation. The classification of tools with the example is provided below in Table 10.

- *RQ3*: Which metrics are used for estimation of usability?

This part gives the details of different metrics and measurement approaches used for estimation of software usability. Table 11 represents metric details with the following primary studies S4, S27, S29, S42, S44–S52, S54–S57, S59, S62, S64, S72, S79, S83, S96, S98, S107, S111, S113, S129, S134, S136, S140, and S147.

In Table 11 it has been mentioned that there are various metrics and measurement approaches that have been adopted for estimation of software usability. The estimation of usability is beneficial because it measures how well

an end-user can interact with a given software product or interactive system. If any software product completely performs all of its technical functions but an end-user cannot get the product to work, then that software product has considered as a failure [191]. Currently, various research practitioners use available standardized usability questionnaires for measuring usability [191, 192]. But few research gaps related to the estimation of usability has been addressed e.g. current methods for measurement of usability in practices do not include all ISO and ANSI defined aspects of usability into a single score or metric. Further, some of the measures are difficult to compute. So, there is a necessity of a method to compute a single comprehensive usability metric by covering all aspect of usability. These are open issues for usability's estimation and research practitioners can pursue in this direction for their future research.

- *RQ4*: Can we integrate usability engineering with software engineering?

This part provides the details for integration of usability related issues during different phases of software development life cycle. Studies which are included in this category are S2–S15, S16–S18, S22–S24, S26, S28, S32(Usability evaluation framework for mass market mobile devices: UTUM), S39, S41–S42, S45, S50, S53,

Table 9 Usability evaluation methods

Methods	Included primary studies	Total number of included primary studies for each approach
Coaching method (CM)	–	0
Codiscovery learning (CL)	–	0
Cooperative usability testing (CUT)	S69, S84, S103 and S123	4
Formal usability testing (FUT)	S32 and S150	2
Journalled sessions(JS)	S140 and S141	2
Log file analysis (LFA)	S150	1
Remote testing (RT)	S76, S87, S92 and S112	4
Retrospective testing (RST)	–	0
Retrospective thinking-aloud (RTA)	–	0
Scenario building (SB)	S99, S106 and S128.	3
Self-reporting logs (SRL)	S141	1
Shadowing method (SM)	–	0
Teaching method (TM)	–	0
Thinking-aloud protocol (TAP)	S26, S31, S35, S57–S58, S75, S93, S125, S135 and S150.	10
Usability testing (UT)	S4, S12, S20, S27, S41, S44, S49, S53, S61, S72–S73, S85–S86, S90, S92, S106, S118, S120, S122–S123, S125–S127, S130–S131, S138, S139 and S150	28
Cognitive walkthrough (CWT)	S27–S28, S33, S75, S93, S95, S114 and S117	8
Consistency inspection (CI)	–	0
Diagnostic evaluation (DE)	–	0
Feature inspection (FI)	S150	1
Formal usability inspection (FUI)	–	0
Groupware walkthrough (GW)	S68	1
Heuristic evaluation (HE)	S3, S12, S30, S36–S37, S40, S44, S62–S63, S65, S79, S86, S88–S90, S114, S117, S120, S122–S123, S126, S138, and S142.	23
Metaphors of human thinking (MHT)	S101 and S129	2
Participatory heuristic evaluation (PHE)	–	0
Participatory walkthrough (PaWT)	–	0
Pluralistic walkthrough (PWT)	–	0
Scenarios-based walkthrough (SB)	S68, S114 and S122	3
Perspective-based inspection (PBI)	–	0
Usability inspection method (UIM)	S17, S28, S30, S71, S91, S119, S133, S142 and S144	9
Contextual inquiry (CI)	S114 and S136	2
Field observation (FO)	S28, S35, S99, S126 and S138	5
Focus groups (FG)	S34, S38, S41, S122, S146 and S149	6
Interviews (I)	S19, S28, S30–S31, S41, S68, S93, S97, S99–S101, S110, S117, S122, S126, S128–S129, S131–S132, S137, S138, S140, S143, S144 and S148	25
Questionnaire (Q)	S16, S21, S27, S32, S34–S35, S38, S72, S83, S90, S96–S97, S99, S101, S115–S116, S122, S126, S128–S129, S133, S135, S136, S138, S140, S141, S142, S144, S146, S147, S148, S149 and S150	33
Screen snapshots (SS)	S141	1
Surveys (S)	S27, S29, S34, S140, S142, S144, S147 and S149	8
User feedback (UF)	S53	1

Table 10 A brief summary of major tools used for usability test lab [190]

Categories of usability tools	Examples of usability tools
Data analysis and mining	SPSS statistical analysis, IBM EZSort, WebQuilt, WebSAT, WebCAT, SAS data mining solution
Observational coding system	Noldus Observer OvoStudio, UsabilityWare, WebVIP and FLUD, Keystroke recording, Camtasia studio, Morae
Online surveys and questionnaires	Customer satisfaction survey, Ezpolls, NetReflector instant survey, Question Master, SNAP survey software, WebSurveyor, Vividence tools
Participant recruiting	PopRecruit
Prototyping and simulation	IBM EasyChart, SILK: Sketching interfaces, SUEDE, CrossWeaver, PatchWork
Remote control, communication and observation	Timbuktu and Netopia, Compaq carbon copy, NetOp for Windows, PCAnywhere, Symantec ProComm Plus Unicenter, Remote Control Control IT
Testing process management and improvement	Observer XT, OvoLogger, Uzilla.
Usability measurement	ErgoLight, Quip, SUMI, WAMMI, MUMMS
Video capture, indexing, storage and retrieval	MPEG-based encoding, IBM VideoAnnEx annotation tool, Adobe Premiere, Screen capture module
Video-conferencing	SightSpeed, Umeeting web conferencing, Microsoft Exchange 2000, conferencing server, Cisco IP Phone, Reality Fusion, TeamView, NetMeetin

Table 11 Metrics used for estimation of usability

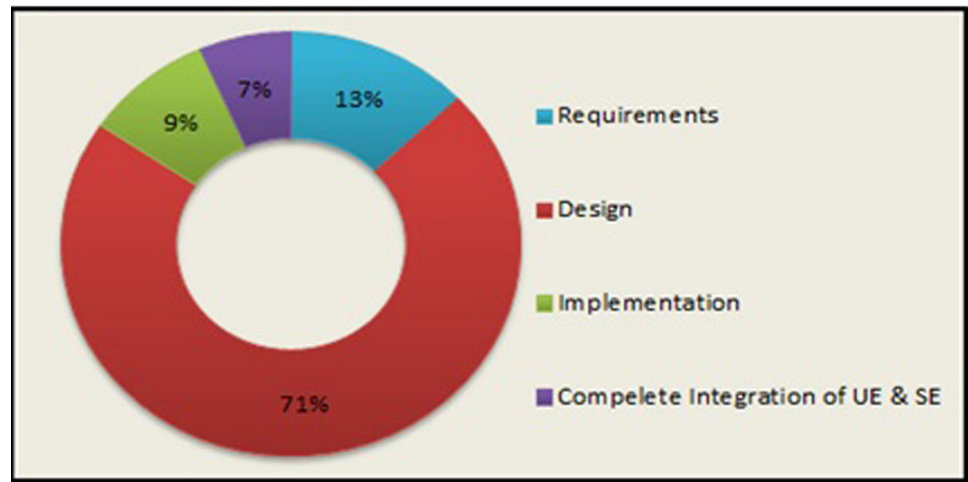
Primary studies	Metrics or measurement approach description
S4	The Empirical measurement is used in which simulation and prototypes are adopted
S27	Usability is assessed by measuring task's completion rate, the elapsed time, number of accesses to help system of tool and instructor's verbal assistance
S29, S72, S83, S134, S140 and S147	SUS: System Usability Scale
S42	Assessment of usability measurements using effort based metrics is done
S45 and S50	Object oriented metric: Ck metric is used to assess usability of software system
S46	SUM: Quantitative model for usability, Six Sigma, Z-score are used
S47	To measure usability, problems with measures are discussed
S48	Measurement approach to quantify attribute are adopted
S66	Static usability measurement in function of effort and time is used
S52	A metric for usability is computed using formulae for three features like access and navigation, orientation and user interaction
S54	A MUS is based on new measurement method of usability (usability magnitude estimation)
S79	Formula based approach is used to compute usability % of a website
S98	Usability Prediction is done using Magnitude Estimation Technique
S96	Usability is rated using a single item and 8-item measure.
S113	Microsoft's mobile usability guidelines are analyzed and defined 10-constructs representing mobile application usability
S129	Mean usability scores are used. Usability assessment is carried out using a combination of recorded measurements

S65, S66, S76–S78, S81–S82, S84, S90, S93, S97–S102, S106, S108, S110, S114–S117, S122–S124, S126–S127, S129, S133, S138–S139, S141–S144, S146, S148 and S150. There is an International standard ISO 13407 which proposes a framework for integration of usability in all

phases of software development process [57, 111]. This standard uses the approach of user-centered design that focuses specifically on developing usability system. Model Driven Engineering [145, 139, 129] is also used as a basis for closer integration of usability practices in the software

Table 12 Phases of SDLC where usability issues are addressed

Phases	Included primary studies
Requirement	S2, S7, S12–S14, S23, S28, S84, S106 and S115
Design	S4–S6, S8–S12, S15–S17, S22–S23, S26, S28, S39–S42, S45, S50, S53, S65–S66, S76, S82, S84, S90, S93, S97–S101, S106, S110, S114–S117, S122–S124, S126, S129, S133, S138–S139, S141–S144, S146, S148 and S150
Implementation	S23, S42, S106, S115, S124, S139 and S150.
Complete integration of UE and SE	S3, S11, S18, S24 and S127

Fig. 6 Phase wise primary studies distribution

development process. Hence, after reviewing above mentioned studies, we conclude that we can integrate usability engineering with software engineering.

- *RQ4.1*: At which phase of software development life cycle, software usability issues are addressed the most?

This section determines the phases of software development life cycle where usability related issues are addressed the most. Table 12. lists different phases with included primary studies.

Out of 150 primary studies, only 77 studies are identified that evaluate usability related issues at different phases of SDLC. Figure 6. represents the phase wise primary studies distribution and also shows that usability issues are mostly addressed during the *design* phase, accounting 71% of included primary studies.

Table 12 represents different the phases of software development life cycle where usability related issues are addressed the most. It can be noticed from Table 12 that research practitioners are aiming to address usability related issues exactly at various stages of software development but their practical implementation is still missing. On the other hand, there is a demand of single precise implemented framework to support the integration of usability engineering with software engineering. In near future,

researchers can work towards this integration with practical implementation.

- *RQ5*: What are the various domains for application of usability related issues?

This section identifies varied domains for applications of software usability related issues. Studies included in this category are S2–S18, S20–S26, S28–S30, S33–S34, S36–S42, S45, S49–S50, S52–S53, S57–S58, S61–S63, S65–S66, S72, S75–S76, S79–S80, S82–S86, S89–S93, S95–S102, S106–S107, S110–S124 and S126–S130. Table 13 presents different domains with included primary studies under the same.

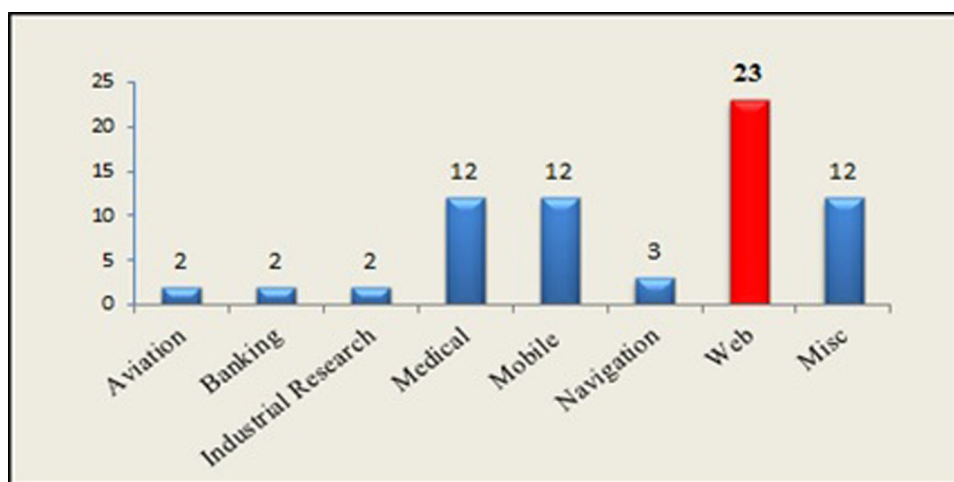
Out of 150 included primary studies, only 68 studies are identified for application of usability into varied domains. Figure 7 represents domain wise primary studies distribution and also shows that 33.82% of potential primary studies use usability evaluation approach in *web-domain*.

6 Limitations

This SLR has considered different primary studies to summarize currently available knowledge of software usability studies. The goal of this review is more focussed

Table 13 Identified domains where usability related issues are addressed

Domains	Included primary studies
Aviation	S123 and S127
Banking	S128 and S129
Industry and research	S32 and S119
Medical	S26, S36–S37, S57, S75, S92, S118, S134 and S136–S139.
Mobile	S15, S25, S29, S41, S72, S90, S101, S113, S116, S121–S122 and S135
Navigation	S16, S21 and S126
Web	S34, S49, S58, S61–S63, S76, S79, S85, S96–S98, S100, S106, S112, S117, S120, S130, S133, S142–S143, S145 and S148.
Miscellaneous	S52, S80, S86, S89, S91, S95, S102, S107, S140–S141, S144 and S149

Fig. 7 Domain wise primary studies distribution

towards identifying and categorizing the relevant primary studies. The study has analyzed the results by considering specific research questions. The primary limitations of this SLR are related to publication bias, misclassification, selection bias and inaccuracy in data extraction by different reviewers in a varied manner. Although, we have searched collectively and exhaustively the entire stated digital search libraries but there may be the possibility that some of the suitable studies may be left out. In general, the review has not considered studies published in national conferences and journals. Furthermore, the review has not considered unpublished results that may pose a threat to this study to some extent. The publication bias has been removed by considering only relevant issues of journals and conference proceedings. The selection bias has been eliminated by defined inclusion and exclusion criteria so that maximum of primary studies can fit into software usability studies. To remove misclassification and inaccuracy in data extraction, 2-authors are considered to conduct classification and extraction of primary studies. All 150 selected primary studies are

classified by each author. The results are computed as applied to reviews published in major international journals and conferences.

7 Discussion and Conclusions

In this paper, we have followed the guidelines and systematic series of steps given by the Kitchenham [41]. We identify 150 primary studies with regards to software usability from 1990 to 2016. Primarily, the study assesses frequently addressed attributes in existing software usability models and standards. The paper recognizes commonly used software usability evaluation methods. The review identifies various metrics used for estimation of usability. The study further finds phases where software usability-related issues are addressed the most. The paper also explores different domains for application of software usability-related issues.

From selected primary studies, the *main findings* are obtained as follows: (a) *Learnability*, *Efficiency*,

Satisfaction, and *Effectiveness* are commonly addressed attributes in different existing software usability models and standards. (b) *Usability Testing*, *Heuristic Evaluation*, and *Questionnaire* are the frequently used methods for evaluation of software usability. (c) *OO*, *ISO* metrics and other measurement approaches¹¹ are used for estimation of software usability. (d) *At Design Phase* of software development, software usability issues are addressed the most as compared to other phases i.e. Requirement and Implementation phase. (e) Usability related issues are evaluated the most for *web-domain*. Other domains, namely, *Aviation*, *Banking*, *Industry and Research*, *Medical*, *Mobile*, and *Navigation Map* are also identified for evaluation of usability related issues.

Here, we present *future guidelines* for research practitioners so that they can carry out future research in software usability engineering:

Various research practitioners have not reached at consensus w.r.t. software usability's models. Hence, there is a need of a consolidated model for usability in non-redundant and consistent manner. Such software usability model would be applied by software developers easily who don't have backgrounds in usability engineering. Developers find confusion for implementation of various methods for evaluation of software usability. So, a single framework is required which facilitates their usage. Current methods for measurement of usability in practices do not include all ISO and ANSI defined aspects of usability into a single score or metric. Furthermore, some of the measures are difficult to compute. So, there is a necessity of a method to compute a single comprehensive usability metric by covering all aspect of usability. Research practitioners must aim to address usability related issues exactly at various stages of software development with their practical implementation. So, there is a demand of single precise implemented framework to support the integration of usability engineering with software engineering. Usability evaluation methods can help in improving electronically delivered services from e-government portals. Intelligent devices, wearable devices, and embedded systems can also be evaluated using usability evaluation methods.

Acknowledgements Darshan Lal, Dipta Sakkarwal, Ricky Tharan, Shruti Singh, Surbhi Garg and Upasana Kundra are gratefully acknowledged for their assistance in providing their suggestions for improvement of this SLR. We also thank Associate Professors of Guru Gobind Singh Indraprastha University for their help in conceptualizing this work.

References

1. Institute of Electrical and Electronics Engineers (1990) IEEE standard glossary of software engineering terminology, IEEE std. 610.12-1990. Author, Los Alamitos
2. ISO 9126 (1991) Information technology-software product evaluation-quality characteristics and guidelines for their use. Geneva
3. Nielsen J (1993) Usability engineering. Academic press, London, ISBN: 978-0-12-518406-9
4. International Organization for Standardization (1998) ISO 9241-11:1998, Ergonomic requirements for office work with visual display terminals (vdt), Part 11: guidance on usability. Author, Geneva
5. Maguire M (2001) Context of use within usability activities. Int J Hum Comput Stud 55:453–483
6. Seffah A, Donyaee M, Kline RB et al (2006) Usability measurement and metrics: a consolidated model. Softw Qual J 14:159–178 [SPRINGER, (S107)]
7. Seffah A, Metzker E (2004) The obstacles and myths of usability and software engineering. Commun ACM 47(12):71–76 (S24)
8. Bertoa MF, Troya JM, Vallecillo A (2005) Measuring the usability of software components. J Syst Softw 79:427–439 (S48)
9. Borgholm T, Madsen KH (1999) Cooperative usability practices. Commun ACM 42(5):91–97 (S103)
10. Graham MJ, Kubose TK, Jordan D et al (2004) Heuristic evaluation of infusion pumps: implications for patient safety in intensive care units. Int J Med Inf 73(11–12):771–779 (S36)
11. Schusteritsch R, Wei CY, Larosa M (2007) Towards the perfect infrastructure for usability testing on mobile devices. In: Proceeding of CHI EA '07 CHI '07 Extended Abstracts on Human Factors in Computing Systems, ACM, pp 1839–1844. (S25)
12. Bodker S, Buur J (2002) The design collaboratorium—a place for usability design. ACM Trans Comput Hum Interact 9(2):152–169 (S15)
13. Clamann M, Kaber BD (2004) Applicability of usability evaluation techniques to aviation systems. Int J Aviat Psychol 14(4):385–420 (S123)
14. Peevers G, Douglas G, Jack AM (2008) A usability comparison of three alternative message formats for an sms banking service. Int J Hum Comput Stud 66:113–123 [ELSEVIER, (S128)]
15. Weir SC, Anderson NJ, Jack AM (2006) On the role of metaphor and language in design of third party payments in eBanking: usability and quality. Int J Hum Comput Stud 64:770–784 [ELSEVIER, (S129)]
16. Chan AJ, Islam MK, Rosewall T et al (2012) Applying usability heuristics to radiotherapy systems. Radiother Oncol 102(1):142–147 [ELSEVIER, (S37)]
17. Kushniruka AW, Triola MM, Borycki EM et al (2005) Technology induced error and usability: the relationship between usability problems and prescription errors when using a handheld application. Int J Med Inf 74:519–526 [ELSEVIER, (S26)]
18. Bastien JMC Usability testing (2010) a review of some methodological and technical aspects of the method. Int J Med Inf 79(4):18–23 [ELSEVIER, (S92)]
19. Mansar LS, Jariwala S, Shahzad M et al (2012) A usability testing experiment for a localized weight loss mobile application. Proc Technol 5:839–848 [ELSEVIER, (S118)]
20. Hochhauser M, Gal E (2015) Weiss LP Negotiation strategy video modeling training for adolescents with autism spectrum disorder: a usability study. Int J Hum Comput Interact 31:472–480 (S146)

¹¹ A complete list of usability metric or measurement approach is provided in Table 13.

21. Kjeldskov J, Skov MB, Stage J (2004) Instant data analysis: conducting usability evaluations in a day. In: Proceeding NordiCHI '04 Proceedings of the third Nordic conference on Human-computer interaction, ACM, pp 233–240 (**S57**)
22. Hua L, Gong Y (2013) Usability evaluation of a voluntary patient safety reporting system: understanding the difference between predicted and observed time values by retrospective think-aloud protocols. *Human-computer interaction*. Springer, New York, pp 94–100 (**S75**)
23. Zapata CB, Fernandez-Aleman LJ, Idri A et al (2015) Empirical studies on usability of mhealth apps: a systematic literature review. *J Med Syst* 39:1 [**Springer, (S134)**]
24. Lewis RJ (2015) Introduction to the special issue on usability and user experience: psychometrics. *Int J Hum Comput Interact* 31:481–483
25. Lewis RJ, Utesch SB (2015) Maher ED Measuring perceived usability: the sus, umux-lite and altusability. *Int J Hum Comput Interact* 31:496–505 (**S147**)
26. Kortum P, Sorber M (2015) Measuring the usability of mobile applications for phones and tablets. *Int J Hum Comput Interact* 31:518–529 (**S29**)
27. Hoehle H, Aljafari R, Ventakatesh V (2016) Leveraging microsoft's mobile usability guidelines: conceptualizing and developing scales for mobile application usability. *Int J Hum Comput Stud* 89:35–53 [**ELSEVIER, (S113)**]
28. Moumane K, Idri A, Abran A (2016) Usability evaluation of mobile applications using iso 9241 and iso 25062 standards. SpringerPlus, 5:1–15 (**S135**)
29. McClellan AM, Karumur PR, Vogel IR et al (2016) Designing an educational website to improve quality of supportive oncology care for women with ovarian cancer: an expert usability review and analysis. *Int J Hum Comput Stud* 32:297–307 (**S139**)
30. Rodriguez DF, Acuna TS, Juristo N (2015) Design and programming patterns for implementing usability functionalities in web applications. *J Syst Softw* 105:107–124 [**ELSEVIER, (S106)**]
31. Bringula PR (2016) Factors affecting web portal information services usability: a canonical correlation analysis. *Int J Hum Comput Interact* 32(10):814–826 (**S148**)
32. Battleson B, Booth A, Weintrop J (2001) Usability testing of an academic library web site: a case study. *J Acad Librariansh* 27(3):188–198
33. Ardito C, Costabile FM, Marsico DM (2006) An approach to usability evaluation of e-learning applications. *Univers Access Inf Soc* 4:270–283
34. Rupp AM, Oppold P, McConnell SD (2015) Evaluating input device usability as a function of task difficulty in a tracking task. *Ergonomics* 58(5):722–735 (**S140**)
35. Jankowski J (2015) Grabowski A Usability evaluation of vr interface for mobile robot teleoperation. *Int J Hum Comput Interact* 31:882–889 (**S20**)
36. Kolade WYA (2015) Integrating usability work into a large inter-organisational agile development project: tactics developed by usability designers. *J Syst Softw* 100:54–66 [**ELSEVIER, (S110)**]
37. Gonzalez MP, Lores J (2008) Granollers A Enhancing usability testing through datamining techniques: a novel approach to detecting usability problem patterns for a context of use. *Inf Softw Technol* 50:547–568 [**ELSEVIER, (S86)**]
38. Kalpna S, Anju S (2016) Enhancing usability inspection through data-mining techniques: an automated approach for detecting usability problem patterns of academic websites. *International conference on intelligent human computer interaction*. Springer, Cham
39. Kalpna S, Anju S (2017) Qualitative usability feature selection with ranking: a novel approach for ranking the identified usability problematic attributes for academic websites using data-mining techniques. *Human-centric computing and information sciences*, Springer, New York [Accepted]
40. Panach IJ, Juristo N, Valverde F et al (2015) A framework to identify primitives that represent usability within model-driven development methods. *Inf Softw Technol* 58:338–354 [**ELSEVIER, (S115)**]
41. Stapić Z (2012) Procedures for performing systematic literature review in software engineering. *CECIIS 2012-23rd International Conference*
42. Fernandez A, Insfran E, Abrahão S (2011) Usability evaluation methods for the web: a systematic mapping study. *Inf Softw Technol* 53:789–817 [**ELSEVIER, (S102)**]
43. Bevan N, Azuma M (1997) Quality in use: incorporating human factors into the software engineering lifecycle. In: *Proceedings of the third IEEE international symposium and forum on software engineering standards*, pp 169–179 (**S1**)
44. Clemmensen T (2011) Templates for cross-cultural and culturally specific usability testing: results from field studies and ethnographic interviewing in three countries. *Int J Hum Comput Int* 27(7):634–669 (**S31**)
45. Heiskari HJ, Kauppinen M, Runonen M et al. (2009) Bridging the gap between usability and requirements engineering. In: *17th IEEE international in requirements engineering conference*, pp 303–308 (**S2**)
46. Winter J, Rönkkö K (2009) Satisfying stakeholders' needs—balancing agile and formal usability test results. *e-Inf Softw Eng J* 3:1 (**S32**)
47. Lallemand C (2011) Toward a closer integration of usability in software development: a study of usability inputs in a model driven engineering process. In: *in proceedings of the 3rd ACM SIGCHI symposium on Engineering interactive computing systems*, pp 299–302 (**S3**)
48. Polgár BP (2015) Using the cognitive walkthrough method in software process improvement. *e-Inf Softw Eng J* 9(1):79–85 (**S33**)
49. Burr J (1999) Bagger K Replacing usability testing with user dialogue. *Commun ACM* 42(5):63–66 (**S4**)
50. Lee Y, Kozar AK (2012) Understanding of website usability: specifying and measuring constructs and their relationships. *Decis Support Syst* 52:450–463 [**ELSEVIER, (S34)**]
51. Bass L (2003) John BE Linking usability to software architecture patterns through general scenarios. *J Syst Softw* 66:187–197 (**S5**)
52. Vilbergsdottira GS, Hvannberg TE, Law ELC (2014) Assessing the reliability, validity and acceptance of a classification-scheme of usability problems (cup). *J Syst Softw* 87:18–37 [**ELSEVIER, (S35)**]
53. Folmer E, Bosch J (2004) Architecting for usability: a survey. *J Syst Softw* 70:61–78 (**S6**)
54. Nebe K, Paelke V (2009) Usability-engineering-requirements as a basis for the integration with software engineering. In: *Proceedings of 13th International Conference Human-Computer Interaction, Part I*, pp 652–659. (**S7**)
55. Folmer E, Gorp JV, Bosch J (2005) Software architecture analysis of usability. In: *Proceedings of international conference eng. human-computer interaction and interactive systems*, pp 38–58 (**S8**)
56. Makri S, Blandford A, Cox LA et al (2011) Evaluating the information behaviour methods:formative evaluations of two methods for assessing the functionality and usability of electronic information resources. *Int J Hum Comput Stud* 69:455–482 [**ELSEVIER, (S38)**]
57. John BE, Bass L, Segura MIS et al. (2005) Bringing usability concerns to the design of software architecture. In: *Proceedings of international conference on engineering human-computer interaction and interactive systems*, pp 1–19 (**S9**)

58. Hiniker A, Sobel K, Hong S, Suh H et al (2016) Hidden symbols: how informal symbolism in digital interfaces disrupts usability for preschoolers. *Int J Hum Comput Stud* 90:53–67 [ELSEVIER, (S39)]
59. Juristo N, Moreno AM, Segura MIS (2007) Analysing the impact of usability on software design. *J Syst Softw* 80(9):1506–1516 (S10)
60. Khawaja AM, Chen F (2014) Marcus N Measuring cognitive load using linguistic features: implications for usability evaluation and adaptive interaction design. *Int J Hum Comput Interact* 30:343–368 (S40)
61. Seffah A, Mohamed T, Mammam HH et al (2008) Reconciling usability and interactive system architecture using patterns. *J Syst Softw* 81:1845–1852 (S11)
62. Aryana B, Clemmensen T (2013) Mobile usability: experiences from iran and turkey. *Int J Hum Comput Interact* 29:220–242 (S41)
63. Juristo N, Moreno AM, Segura MS (2007) Guidelines for eliciting usability functionalities. *IEEE Trans Softw Eng* 33(11):744–758 (S12)
64. Tamir DE, Mueller CJ (2010) Pinpointing usability issues using an effort based framework. In: IEEE, international conference on systems man and cybernetics, pp 931–938 (S42)
65. Seffah A, Djouab R, Antunes H (2001) Comparing and reconciling usability-centered and use case-driven requirements engineering processes. *Aust Comput Sci Commun IEEE Comput Soc* 23(5):132–139 (S13)
66. Moorthy JTS, Ibrahim SB, Mahrin MN (2013) Formulation of usability risk assessment model, IEEE, Conference on open systems (S43)
67. Beu A, Honold P (2000) Yuan X How to build up an infrastructure for intercultural usability engineering. *Int J Hum Comput Interact* 12(3 and 4):347–358 (S14)
68. Nielsen J, Landauer KT (1993) A mathematical model of the finding of usability problems, ACM, Conference on Human Factors in Computing Systems pp 206–213 (S44)
69. Dubey KS, Rana A (2011) Usability estimation of software system by using object-oriented metrics. *ACM SIGSOFT Softw Eng Notes* 36(2):371–382 (S45)
70. Roberts JM, Newton JE, Lagattolla DF et al (2013) Objective versus subjective measures of paris metro map usability: investigating traditional octolinear versus all-curves schematics. *Int J Hum Comput Stud* 71:363–386 [ELSEVIER, (S16)]
71. Sauro J, Kindlund E (2005) A method to standardize usability metrics into a single score, ACM, CHI. 1-58113-998-5 (S46)
72. Hertzum M (2006) Problem prioritization in usability evaluation: from severity assessments toward impact on design. *Int J Hum Comput Interact* 21(2):125–146 (S17)
73. Hornbaek K (2006) Current practice in measuring usability: challenges to usability studies and research. *Int J Hum Comput Stud* 64:79–102 (S47)
74. Metzker M, Offergeld M (2001) An interdisciplinary approach for successfully integrating human-centered design methods into development processes practiced by industrial software development organizations. In: Proceedings of International Conference on Engineering for Human-Computer Interaction. Springer, New York (S18)
75. Hertzum MA, Clemmensen T (2012) How do usability professionals construe usability? *Int J Hum Comput Stud* 70:26–42 [ELSEVIER, (S19)]
76. Erickson W, Trerise S, Lee C et al (2013) The accessibility and usability of college websites: is your website presenting barriers to potential students? *Commun Coll J Res Pract* 37:864–876 (S49)
77. Dubey KS, Rana A (2010) Assessment of usability metrics for object-oriented software system. *ACM SIGSOFT Softw Eng Notes* 35 (S50)
78. Lavie T, Gilad OT (2011) Meyer J Aesthetics and usability of in-vehicle navigation displays. *Int J Hum Comput Stud* 69:80–99 [ELSEVIER, (S21)]
79. Frojkaer E, Hertzum M, Hornbaek K (2000) Measuring usability: are effectiveness, efficiency and satisfaction really correlated, ACM Press, CHI, pp 345–352 (S51)
80. Carvajal L, Moreno AM, Segura MIS et al (2013) Usability through software design. *IEEE Trans Softw Eng* 39(11):1582–1596 (S22)
81. Babiker EM, FH Boyle et al. (1991) A metric for hypertext usability, ACM Press. In: Proceedings of 11th annual international conference on system documentation, pp 95–104 (S52)
82. Carvajal L (,) Usability-enabling guidelines: a design pattern and software plug-in solution. In: ESEC/FSE Doctoral symposium, ACM (S23)
83. Beckles B, Welch V (2005) Basney J Mechanisms for increasing the usability of grid security. *Int J Hum Comput Stud* 63:74–101 [ELSEVIER, (S53)]
84. McGee M (2004) Master usability scaling: magnitude estimation and master scaling applied to usability measurement, ACM. In: Proceedings of CHI, pp 335–342 (S54)
85. Nielsen J, Levy J (1994) Measuring usability: preference vs. performance. *Commun ACM* 37:66–76 (S55)
86. Sauro J, Kindlund E (2005) Making sense of usability metrics: usability and six sigma, UPA conference (S56)
87. Teruel AM, Navarro E, Jaquero LV et al (2014) A cscw requirements engineering case tool: development and usability evaluation. *Inf Softw Technol* 56:922–949 [ELSEVIER, (S27)]
88. Lindgaard G (2015) Challenges to assessing usability in the wild: a case study. *Int J Hum Comput Interact* 31:618–631 (S28)
89. Skov MB, Stage J (2005) Supporting problem identification in usability evaluation, CHI. In: Proceedings of OZCHI. 1-59593-222-4. [S58]
90. Mapayi T, Olaniyan OM, Isamotu NO et al (2013) Evaluating usability factors in different authentication methods using artificial neural network. *IEEE. Afr J Comput ICT* 6(1):69–78 (S59)
91. Johannessen JHG, Hornbæk K (2014) Must evaluation methods be about usability? devising and assessing the utility inspection method. *Behav Inf Technol* 33(2):195–206 (S30)
92. Ivory MY, Hearst MA (2001) The state of the art in automated usability evaluation of user interfaces. *ACM Comput Surv* 33:470–516 (S60)
93. Lee S, Cho JE (2007) Usability evaluation of korean e-government portal, SPRINGER. In: universal access in human-computer interaction applications and services, pp 64–72 (S61)
94. González M, Masip L, Granollers A et al (2009) Quantitative analysis in a heuristic evaluation experiment. *Adv Eng Softw* 40:1271–1278 [Elsevier, (S62)]
95. Delice EK (2009) Z Gungor The usability analysis with heuristics evaluation and analytic hierarchy process. *Int J Ind Ergon* 39:934–993 [Elsevier, (S63)]
96. Ríos AD, García VA, Rey ME et al (2010) Usability: a critical analysis and taxonomy. *Int J Hum Comput Interact* 26(1):53–74 (S108)
97. Bradner E, Dawe M (2008) Parts of the sum: a case study of usability benchmarking using the sum metric. In: UPA international conference (S64)
98. Winter S, Wagner S, Deissenboeck F (2008) A comprehensive model of usability, Springer, International Federation for Information Processing, pp 106–122 (S109)
99. Kline RB, Seffah A (2005) Evaluation of integrated software development environments: challenges and results from three empirical studies. *Int J Hum Comput Stud* 63:607–627 [ELSEVIER, (S65)]

100. Tamir D, Komogortsev OV, Mueller CJ (2008) An effort and time based measure of usability, ACM, WoSQ. 978-1-60558-023-4 (**S66**)
101. Scheller T, Kuhn E (2015) Automated measurement of api usability: the api concepts framework. Inf Softw Technol 61:145–162 [**ELSEVIER, (S111)**]
102. Tullis T, Fleischman S, McNulty M et al. (2002) An empirical comparison of lab and remote usability testing of web sites, In: Proceedings of Usability Professionals Conference (**S67**)
103. Santana VFD, Baranauskas MCC (2015) Welfit: a remote evaluation tool for identifying web usage patterns through client-side logging. Int J Hum Comput Stud 76:40–49 [**ELSEVIER, (S112)**]
104. Følstad A (2007) Work-domain experts as evaluators: usability inspection of domain-specific work support systems. Int J Hum Comput Interact 22(3):217–245 (**S68**)
105. Frøkjær E, Hornbæk K (2005) Cooperative usability testing: complementing usability tests with user-supported interpretation sessions. In: CHI' 05 extended abstracts on human factors in computing systems, ACM, pp 1383–1386 (**S69**)
106. Brown M, Sharples S, Harding J (2013) Introducing pegi: a usability process for the practical evaluation of geographic information. Int J Hum Comput Interact 71:668–678 [**ELSEVIER, (S114)**]
107. Thompson KE, Rozanski EP, Haake AR (2004) Here, there, anywhere: remote usability testing that works in, SIGITE, ACM, New York, pp 132–137 (**S70**)
108. Bak JO, Nguyen K, Risgaard P et al (2008) Obstacles to usability evaluation in practice: a survey of software development organizations, ACM. In: Proceedings of Nordi CHI (**S71**)
109. Harrison R, Flood D, Duce D (2013) Usability of mobile applications: literature review and rationale for a new usability model. J Interact Sci 1:1
110. Winter J, Hinely M (2011) Examining correlations in usability data to effectivize usability testing. e-Inf Softw Eng J 5(1):25–37 (**S72**)
111. Castilla D, Palacios GA, Lopez BJ et al (2013) Process of design and usability evaluation of a telepsychology web and virtual reality system for elderly: butler. Int J Hum Comput Stud 71:350–362 [**ELSEVIER, (S117)**]
112. Bruun A, Stage J (2012) Training software development practitioners in usability testing: an assessment acceptance and prioritization, OZCHI, ACM (**S73**)
113. Sauer J, Seibel K, Ruttinger B (2014) The influence of user expertise and prototype fidelity in usability tests. Appl Ergon 41:130–140 [**ELSEVIER, (S74)**]
114. Hurtado N, Ruiz M, Orta E et al (2015) Using simulation to aid decision making in managing the usability evaluation process. Inf Softw Technol 57:509–526 (**S119**)
115. Hasan L, Morris A, Proberts S (2012) A comparison of usability evaluation methods for evaluating e-commerce websites. Behav Inf Technol 31(7):707–737 (**S120**)
116. Hong JI, Heer J, Waterson S et al (2001) Webquilt: a proxy-based approach to remote web usability testing. ACM Trans Inf Syst 19(3):263–285 (**S76**)
117. Kjeldskov J, Stage J (2004) New techniques for usability evaluation of mobile systems. Int J Hum Comput Stud 60:599–620 [**ELSEVIER, (S121)**]
118. Nielsen J, Molich R (1990) Heuristic evaluation of user interfaces. In: Proceedings ACM CHI pp 249–256 (**S77**)
119. Ji GY, Park HJ, Lee C et al (2006) A usability checklist for the usability evaluation of mobile phone user interface. Int J Hum Comput Interact 3(3):207–231 (**S122**)
120. Nielsen J (1994) Enhancing the explanatory power of usability heuristics, ACM, CHI Human factors in computing systems, 0-89791-650-6/94/0152 (**S78**)
121. Torrente SCM, Prieto MBA, Gutiérrez AD et al (2013) Sirius: a heuristic-based framework for measuring web usability adapted to the type of website. J Syst Softw 86:649–663 [**ELSEVIER, (S79)**]
122. Folmer E, Welie VM, Bosch J (2006) Bridging patterns: an approach to bridge gaps between se and hci. Inf Softw Technol 48:69–89 [**ELSEVIER, (S124)**]
123. Andreasen MS, Nielsen HV, Schrøder SO (2007) What happened to remote usability testing? an empirical study of three methods, CHI, ACM, pp 1405–1414. (**S80**)
124. Roberts LV, Fels ID (2006) Methods for inclusion: employing think aloud protocols in software usability studies with individuals who are deaf. Int J Hum Comput Stud 64:489–501 [**ELSEVIER, (S125)**]
125. Wright PC, Monk AF (1991) The use of think—aloud evaluation methods in design. ACM SIGCHI 23(1):55–57 (**S81**)
126. Nivala MA, Sarjakoski TL, Sarjakoski T (2007) Usability methods familiarity among map application developers. Int J Hum Comput Stud 65:784–795 (**S126**)
127. Wharton C, Rieman J, Lewis C et al (1994) The cognitive walkthrough method: a practitioner's guide. In: Nielsen J, Mack RL (eds) Usability inspection methods. Wiley, Hoboken, pp 105–140 (**S82**)
128. Propp S, Buchholz G, Forbrig P (2009) Integration of usability evaluation and model-based software development. Adv Eng Softw 40:1223–1230 [**ELSEVIER, (S127)**]
129. Leow CM, Wang KYL, Lau HS et al (2016) Usability of rpg-based learning framework. Int J Hum Comput Interact. doi:10.1080/10447318.2016.1183863 (**S83**)
130. Følstada A, Hornbæk K (2010) Work-domain knowledge in usability evaluation: experiences with cooperative usability testing. J Syst Softw 83(11):2019–2030 (**S84**)
131. Skov MB, Stage J (2004) Integrating usability design and evaluation: training novice evaluators in usability testing. In: Proceedings of the workshop on Improving the interplay between Usability Evaluation and User Interface Design, ACM (**S85**)
132. Isa WAM, Suhani MR, Safie NI et al (2011) Accessing the usability and accessibility of malaysia e-government website. Am J Econ Bus Adm 3(1):40–46 (**S130**)
133. Yammiyavar P, Clemmensen T (2008) Kumar J Influence of cultural background on non-verbal communication in a usability testing situation. Int J Des 2(2):31–40 (**S131**)
134. Hartson HR, Castillo JC, Kelso J (1996) Remote evaluation: the network as an extension of usability laboratory, ACM, CHI. 089791-777-4/96/04 (**S87**)
135. Hertzum M, Clemmensen T, Hornbæk K et al (2011) Personal usability constructs: how people construe usability across nationalities and stakeholder groups. Int J Hum Comput Interact 27(8):729–761 (**S132**)
136. Nielsen J (1992) Finding usability problems through heuristics evaluation. In: Proceedings ACM CHI pp 373–380 (**S88**)
137. Rivero L, Conte T (2013) Using an empirical study to evaluate the feasibility of a new usability inspection technique for paper based prototypes of web applications. J Softw Eng Res Dev 1:2 [**Springer, (S133)**]
138. Mankoff J, Dey AK, Hsieh G et al (2003) Heuristic evaluation of ambient displays, ACM, CHI: new horizons, 5:1 (**S89**)
139. Ko MS, Chang SW, Ji GY (2013) Usability principles for augmented reality applications in a smartphone environment. Int J Hum Comput Interact 29:501–515 (**S90**)
140. Sawyer P, Flanders A, Wixon D (1996) Making a difference—the impact of inspections. In: Proceedings of CHI, ACM press (**S91**)
141. Hamborg CK, Vehse B, Bludau BH (2004) Questionnaire based usability evaluation of hospital information systems. Electron J Inf Syst Eval 7(1):21–30 (**S136**)

142. Arnhold M, Quade M, Kirch W (2014) Mobile applications for diabetics: a systematic review and expert-based usability evaluation considering the special requirements of diabetes patients age 50 years or older. *J Med Internet Res* 14(4):1–18 (**S137**)
143. Paradowski M, Fletcher A (2004) Using task analysis to improve usability of fatigue modelling software. *Int J Hum Comput Stud* 60:101–115 [**ELSEVIER, (S93)**]
144. Mirkovic J, Kaufman RD (2014) Rutland MC (2014) Supporting cancer patients in illness management: usability evaluation of a mobile app. *JMIR MHEALTH AND UHEALTH* 2(3):1–21 (**S138**)
145. Holzinger A (2005) Usability engineering methods for software developers. *Commun ACM* 48(1):71–74 (**S94**)
146. Spencer R (2000) The streamlined cognitive walkthrough method: working around social constraints encountered in software development company. In: *Proceedings of the conference on human factors in computing systems (CHI)*, ACM press, New York (**S95**)
147. Christophersen T (2011) Konradi U Reliability, validity and sensitivity of a single-item measure of online store usability. *Int J Hum Comput Stud* 69:269–280 [**ELSEVIER, (S96)**]
148. Datcu D, Lukosch S, Brazier F (2015) On the usability and effectiveness of different interaction types in augmented reality. *Int J Hum Comput Interact* 31:193–209 (**S141**)
149. Leuthold S, Avila BAJ, K K (2008) Beyond web content accessibility guidelines: design of enhanced text user interfaces for blind internet users. *Int J Hum Comput Stud* 66:257–270 [**ELSEVIER, (S97)**]
150. Bias GR, Moon MB (2015) Hoffman RR Concept mapping usability evaluation: an exploratory study of a new usability inspection method. *Int J Hum Comput Interact* 31(9):571–583 (**S142**)
151. Kim JK, Han HS, Yun HM et al (2002) A systematic procedure for modeling usability based on product design variables: a case study in audiovisual consumer electronic products. *Int J Occup Saf Ergon* 8(3):387–406 (**S98**)
152. Tung LL, Xu Y, Tan BF (2009) Attributes of web site usability: a study of web users with the repertory grid technique. *Int J Electron Commer* 13(4):97–126 (**S143**)
153. Chung W, Chen H, Chaboya GL et al (2005) Evaluating event visualization: a usability study of coplink spatio-temporal visualizer. *Int J Hum Comput Stud* 62:127–157 [**ELSEVIER, (S99)**]
154. Choi KI, Kim SW, Lee D et al (2015) A weighted qfd-based usability evaluation method for elderly in smart cars. *Int J Hum Comput Interact* 31(10):703–716 (**S144**)
155. Roth V, Straub T, Richter K (2005) Security and usability engineering with particular attention to electronic mail. *Int J Hum Comput Stud* 63:51–73 [**ELSEVIER, (S100)**]
156. Nawaz A (2013) Clemmensen T Website usability in asia”from within: an overview of a decade of literature. *Int J Hum Comput Interact* 29:256–273 (**S145**)
157. Howell M, Love S, Turner M (2006) Visualisation improves the usability of voice-operated mobile phone services. *Int J Hum Comput Stud* 64:754–769 [**ELSEVIER, (S101)**]
158. West R, Lehman KR (2006) Automated summative usability studies: an empirical evaluation, ACM, CHI 2006 Proceedings. 1-59593-178-3/06/0004 (**S104**)
159. Shamim A, Balakrishnan V, Tahir M et al (2016) Age and domain specific usability analysis of opinion visualisation techniques. *Behav Inf Technol* 35(8):680–689 (**S149**)
160. Wixon D (2003) Evaluating usability methods: why the current literature fails the practitioner, ACM, New York, 1072-5220/03/0700 (**S105**)
161. Hanrath S, Kottman M (2015) Use and usability of a discovery tool in an academic library. *J Web Librariansh* 9:1–21 (**S150**)
162. Abran A, Khelifi A, Suryn W (2003) Usability meanings and interpretations in iso standards. *Softw Qual J* 11:325–338
163. Arms WY (2000) Digital libraries. MIT Pr, Cambridge
164. Bevan N, Kirakowski J, Maissel J (1991) What is usability? In: *Proceedings of the 4th international conference on HCI*, pp 651–655
165. Boëhm B (1978) Characteristics of software quality, Vol 1 TRW series on software technology. North-Holland, Amsterdam
166. Booth P (1989) An introduction to human-computer interaction. Lawrence Erlbaum Associates Publishers, Hove/East Sussex
167. Lewis RJ (1995) IBM computer usability satisfaction questionnaires: psychometric evaluation and instructions for use. *Int J Hum Comput Interact* 7:57–78
168. Blandford A, George B (2002) Usability for digital libraries. In: *Proceedings of the second ACM/IEEECS joint conference on digital libraries*. ACM Press, New York, p 424
169. Brinck Tom, Gergle Darren, Wood Scott D (2002) Designing web sites that work: usability for the web. Morgan Kaufmann, San Francisco
170. Constantine L, Lockwood LAD (1999) Software for use: a practical guide to the models and methods of usage-centered design. Addison-Wesley, New York
171. Dix A, Finley J, Abowd G et al (1998) Human-computer interaction, 2nd edn. Prentice-Hall, Englewood Cliffs
172. Donyaee M, Seffah A (2001) Quim: an integrated model for specifying and measuring quality in use. In: *Eighth IFIP Conference on Human Computer Interaction*. Tokyo, Japan
173. Grady RB (1992) Practical software metrics for project management and process improvement. Prentice Hall, Englewood Cliffs
174. Gould JD (1988) How to design usable systems. In: Helander M (ed) *Handbook of human computer interaction*. Elsevier, New York, pp 57–89
175. Hix D, Hartson HR (1993) Developing user interfaces: ensuring usability through product & process. Wiley, New York
176. Löwgren J (1993) Human-computer interaction: What every system developer should know. Studentlitteratur, Lund
177. McCall JA, Richards PK, Walters GF (1977) Factors in software quality, vol II. Rome Aid Defence Centre, Italy
178. Macleod M, Bowden R, Bevan N et al (1997) The music performance method. *Behav Inf Technol* 16:279–293
179. Preece J, Benyon D, Davies G et al (1993) A guide to usability: human factors in computing. Addison-Wesley, Reading
180. Preece J, Rogers Y, Sharp H et al (1994) Human-computer interaction. Addison-Wesley, Reading
181. Quesenberry W (2001) What does usability mean: looking beyond ease of use. In: *Proceedings of the 18th Annual Conference Society for Technical Communications*
182. Shneiderman B, Plaisant C (2005) Designing the user interface: strategies for effective human-computer interaction. Addison Wesley, Boston
183. Shackel B (1991) Usability—context, framework, definition, design and evaluation. In: Shackel B, Richardson SJ (eds) *Human factors for informatics usability*. Cambridge University Press, New York, pp 21–37
184. Bevan N (1995) Human-computer interaction standards. In: *Proceedings of the 6th international conference on human computer interaction*, Elsevier, Amsterdam
185. Bevan N (2009) International standards for usability should be more widely used. *J Usability Stud* 4(3):106–113
186. Bevan N, Schoeffel R (2001) A proposed standard for consumer product usability. In: *Proceedings of 1st international conference on universal access in human computer interaction*, pp 557–561

187. DIN EN ISO 13407 (1999s) Human-centered design processes for interactive systems, CEN—European Committee for Standardization, Brussels
188. ISO/PAS 18152 (2003) Ergonomics of human-system interaction-specification for the process assessment of human-system issues. ISO, Genf
189. Lewis RJ (2014) Usability: lessons learned...and yet to be learned. *Int J Hum Comput Interact* 30:663–684
190. Seffah A, Mammarr HH (2009) Usability engineering laboratories: limitations and challenges toward a unifying tools/practices environment. *Behav Inf Technol* 28(3):281–291 **(94)**
191. Kortum Philip T, Bangor Aaron (2013) Usability ratings for everyday products measured with the System Usability Scale. *Int J Hum Comput Interact* 29(2):67–76
192. Kortum P, Peres CS (2014) The relationship between system effectiveness and subjective usability scores using the system usability scale. *Int J Hum Comput Interact* 30:575–584